

VotaSJ

**Digital participatory budgeting for São José/SC — verifiable by anyone,
trusted by no one alone**

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INE5458 — Blockchain and Cryptocurrency Technologies · UFSC 2026/1

`github.com/daviludvig/votasj-dapp`

"A blockchain-anchored participatory budget where every citizen votes from their phone, and every vote can be independently recomputed by anyone — not just trusted."

-  Two Solidity contracts, 100% test coverage
-  One command reproduces the whole cycle: `npm run demo`
-  Built for Polygon PoS — same code, local, testnet, or mainnet

< 2% turnout

180,000 eligible voters in São José/SC.

Fewer than 3,600 decide how tens of millions of reais are spent.

- BRL 300–500k per cycle in physical-process cost (assemblies, poll workers, ~3-week manual tally)
- The City Hall counts its own vote — every contested result ends in an unfalsifiable fraud accusation

BRL 60M+/yr addressable

Tier	Voters	Revenue/yr
São José/SC (anchor)	~180k	~BRL 220k
Greater Florianópolis (4 cities)	~1.1M	BRL 0.8–1.2M
Santa Catarina state	~5.5M	BRL 4–6M
Brazil, cities >50k inhabitants	~85M	BRL 60M+

Five steps, one command

```
register → open cycle → propose → vote → close & tally
```

```
npm run demo
```

Every step prints its transaction hash, gas cost, and decoded event. Five illegal actions are attempted on purpose — and rejected, on camera.

Live demo follows.

Team

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100% coverage, 43 tests, 0 hand-waving

- 43 automated tests, 100% statement/branch/function/line coverage — plus one test at **500 independent voters**, tally still checks out and gas per vote stays flat
- `npm run demo`: a self-checking, reproducible run that tries 5 illegal actions and asserts every one reverts
- CI runs lint + test + coverage + metrics + demo on every push — every report is a downloadable build artifact

BRL 170–300k saved per cycle

- Annual subscription: BRL 180–250k / municipality
- Per-cycle setup fee: BRL 40k
- Existing process: BRL 300–500k/cycle → VotaSJ, all-in: BRL 90–160k/cycle
- **Citizens never pay**

What's proven vs what's next

Now (PD3, delivered) — VoterRegistry + ParticipatoryBudget , 100% coverage, reproducible demo, CI-enforced

Next — gov.br identity relay · IPFS proposal pinning · ExecutionTracker for milestone accountability

Then — Public Polygon Amoy deployment, verified on Polygonscan · Citizen-facing PWA

Later — 3-of-5 multisig · 14-day timelock · zk vote privacy

Don't trust the count — recompute it

Ask: an institutional pilot conversation with São José/SC and a BRL 400–600k seed envelope for the external audit, gov.br certification, and the field pilot.

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